



A comprehensive experimental and modeling study of iso-pentanol combustion



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ABSTRACT

Biofuels are considered as potentially attractive alternative fuels that can reduce greenhouse gas and pollutant emissions. iso-Pentanol is one of several next-generation biofuels that can be used as an alternative fuel in combustion engines. In the present study, new experimental data for iso-pentanol in shock tube, rapid compression machine, jet stirred reactor, and counterflow diffusion flame are presented. Shock tube ignition delay times were measured for iso-pentanol/air mixtures at three equivalence ratios, temperatures ranging from 819 to 1252 K, and at nominal pressures near 40 and 60 bar. Jet stirred reactor experiments are reported at 5 atm and five equivalence ratios. Rapid compression machine ignition delay data was obtained near 40 bar, for three equivalence ratios, and temperatures below 800 K. Laminar flame speed data and non-premixed extinction strain rates were obtained using the counterflow configuration. A detailed chemical kinetic model for iso-pentanol oxidation was developed including high- and low-temperature chemistry for a better understanding of the combustion characteristics of higher alcohols. First, bond dissociation energies were calculated using *ab initio* methods, and the proposed rate constants were based on a previously presented model for butanol isomers and *n*-pentanol. The model was validated against new and existing experimental data at pressures of 1–60 atm, temperatures of 650–1500 K, equivalence ratios of 0.25–4.0, and covering both premixed and non-premixed environments. The method of direct relation graph (DRG) with expert knowledge (DRGX) was employed to eliminate unimportant species and reactions in the detailed mechanism, and the resulting skeletal mechanism was used to predict non-premixed flames. In addition, reaction path and temperature A-factor sensitivity analyses were conducted for identifying key reactions at various combustion conditions.

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1. Introduction

The interest in alternative fuels and fuel additives has increased in recent years. Oxygenated fuels have been considered as alternative fuels in order to reduce NO_x and particulate emissions. In addition, the production of oxygenated fuels from renewable sources can balance emissions of the major green-

house gas (CO₂) from combustion devices. However, the fundamental combustion properties of oxygenated fuels need to be evaluated prior to application [1]. Ethanol is an attractive alternative bio-based alcohol fuel extender for petroleum fuels. Although ethanol can reduce the dependency upon petroleum fuels and thus greenhouse gas emissions, disadvantages such as high O/C ratio, high hygroscopicity, and low energy density can cause problems with fuel storage, blending, and use in engines. On the other hand, higher alcohols have lower hygroscopicity and corrosivity, higher energy density, and can be readily blended with hydrocarbon fuels in fueling systems.

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Recently, the interest for using iso-pentanol (3-methyl-1-butanol, isoamyl alcohol) as a bio-derived [2] gasoline substitute has increased because it has a higher energy density and is less hygroscopic than ethanol. However, there have been limited studies on iso-pentanol combustion. Yang et al. studied the fundamental characteristics of iso-pentanol as a fuel for HCCI engines [3], and the results were also modeled by Tsujimura and co-workers using a detailed chemical kinetic model [4]. These HCCI engine studies indicate that iso-pentanol displays intermediate temperature heat release (ITHR) characteristics, which is important for optimal engine operation. Dayma et al. [5] measured the concentrations of reactants, intermediate species, and products over a range of equivalence ratios and temperatures in a jet stirred reactor (JSR) at 10 atm, and simulated their experiments using a detailed chemical kinetic model. Recently, Tsujimura et al. [6] proposed a chemical kinetic model for iso-pentanol and validated it against ignition delay times measured in a shock tube (ST) at high temperatures and a rapid compression machine (RCM) at low temperatures. This was the first study to include comprehensive reaction pathways to predict iso-pentanol ignition behavior at low, intermediate, and high temperatures. Tang et al. measured the high temperature ignition behavior of iso-pentanol [7] and found good predictions using the model proposed by Dayma [5]. Their experimental results show that iso-pentanol is less reactive than *n*-pentanol. Zhao and co-workers have calculated pressure and temperature dependent rate constants for the thermal decomposition of iso-pentanol [8]. Welz and co-workers [9] investigated the low temperature oxidation mechanism of iso-pentanol using photoionization mass spectrometry. Their results indicate that the formation of iso-pentanal (3-methylbutanal) and enols are important, whereas cyclic ether channels are not. More recently, Welz and co-workers [10] used quantum chemical theory together with experiments to characterize an unconventional water elimination pathway from hydroxypentylhydroperoxide (i.e., QOOH) radicals derived from iso-pentanol, which they suggest would alter alcohol oxidation mechanisms.

The purpose of the current study is to expand the body of experimental data available for iso-pentanol combustion, and to improve the mechanistic understanding of its combustion using detailed chemical kinetic modeling. We present a comprehensive dataset for iso-pentanol oxidation spanning a wide range of temperatures, pressures, equivalence ratios, and experimental configurations. A comprehensive chemical kinetic model for iso-pentanol is developed using rules established for modeling other C_4 and C_5 alcohols together with new and existing [8,10] quantum chemical calculations. The model is validated against recently published experimental data as well as new data presented herein.

2. Chemical kinetic model formulation

The detailed chemical kinetic model for iso-pentanol includes both low- and high-temperature chemistry. The proposed model is based on previous modeling study on the combustion of the C_4 and C_5 alcohols [11,12], and a similar methodology was used in this study to develop a detailed model for iso-pentanol. The present model distinguishes itself from previous iso-pentanol models available in the literature in the following ways:

- The model developed by Dayma et al. [5] for predicting JSR data did not require low-temperature chemistry because the experiments did not display low temperature reactivity. The present work includes low temperature chemistry, which is needed to predict ST and RCM ignition delay presented herein.

- The model developed by Tsujimura et al. [6] includes both low- and high-temperature chemistry. The present model follows a similar philosophy; however, we restructured the model following recent work on C_4 and C_5 alcohols [11,12] and updated all the reaction classes and rate rules according to recent fundamental iso-pentanol kinetic studies, as discussed in detail below.
- The present model includes transport properties for species in the high-temperature mechanism, which is a necessity for flame simulations.
- The present model is reduced using a computer algorithm to simulate counterflow flame extinction.

Only a brief description of the model development process is discussed herein, and readers are referred to [11] and its [Supplementary material](#) for a more detailed description of the methods employed. The thermodynamic data for iso-pentanol and related radicals were calculated using the THERM program of Ritter and Bozzelli [13]. The detailed chemical kinetic mechanism utilized in this work is based on the hierarchical nature of combustion mechanisms. Therefore, the reaction mechanism of iso-pentanol was developed by adding its primary reactions and related radical reactions to the *n*-pentanol [12] and iso-butanol [11] reaction mechanisms. The chemistry of C_5 aldehydes and enols was developed following the methods described in detail for C_4 aldehydes and enols in [11,14]. All the validation simulations were conducted in CHEMKIN PRO 15112 [15] using the appropriate reactor modules. The entire model consists of 360 species and 2075 reactions, and the CHEMKIN input files are available as [Supplementary material](#) to this publication and on the website <http://cpc.kaust.edu.sa>.

2.1. Quantum chemistry calculations

The reaction rates selected in this study follow directly from previous work on alkanes and alcohols [11,16–19]. This approach was initially validated using quantum chemistry calculations to verify the most stable iso-pentanol conformation and to obtain C–H and C–C bond dissociation energies (BDE). All geometry optimizations and frequency calculations are performed using the Gaussian09 suite of programs [20]. The G3, G4 and CBS-QB3 composite methods are selected for their reported accuracies of 1.0, 1.0 and 1.1 kcal mol^{−1}, respectively [21–23]. The conformational analysis of iso-pentanol and the related bond dissociation products is conducted to determine the most stable configuration of the reactant and product species. Similar to previous studies on *n*-butanol [24] and *n*-pentanol [12] the conformations of iso-pentanol will be denoted with a G⁺, G[−] or T for +60°, −60° or 180° dihedral angles, respectively. Since Moc et al. demonstrated that for a primary OH group the trans CCOH is most stable, by 0.3 kcal mol^{−1}, it is the only arrangement used in this study and will be designated by a “t” [24]. In the conformer naming, the designation for the dihedral corresponding to the branched methyl groups will be presented first and will be separated from the other labels of the other subsequent dihedrals by an underscore. For each conformer, both iso-pentanol and the products of C–X and X–H bond scissions are analyzed.

Table 1 gives the relative stabilities of the various conformers of iso-pentanol and the BDE for the most stable product configuration. A complete table comparing all of the product configurations is available in the [Supplementary material](#) (Table S1). The G4 BDE values are consistently 2.1 ± 0.2 kcal mol^{−1} lower than both the CBS-QB3 and G3 values. Although these differences are significantly larger than the reported accuracies of each method, the stabilities of each conformer and the trends in the BDE values are in excellent agreement, varying by 0.1–0.2 kcal mol^{−1}. The CBS-QB3 and G3 values are more consistent with experimental results for

Table 1

Difference in enthalpy between the conformers of iso-pentanol and the bond dissociation energies of the lowest energy conformer. Energies are relative to the $G^+_{TG^-t}$ conformation of iso-pentanol (kcal mol^{-1}).

Reaction	Conformer	CBS-QB3		G3		G4	
		ΔH (0 K)	ΔH (298 K)	ΔH (0 K)	ΔH (298 K)	ΔH (0 K)	ΔH (298 K)
Iso-pentanol	$G^+_{TG^-t}$	0.0	0.0	0.0	0.0	0.0	0.0
	$G^+_{TG^+t}$	1.2	1.2	1.2	1.2	1.3	1.2
	G^-_{TTt}	0.6	0.6	0.5	0.6	0.5	0.5
	$G^-_{G^+G^-t}$	1.9	1.9	1.8	1.8	2.0	1.9
	$G^-_{G^+G^+t}$	1.9	1.8	1.8	1.8	1.9	1.9
	$G^-_{G^+Tt}$	1.6	1.6	1.5	1.6	1.5	1.5
	$T_{G^-G^-t}$	1.2	1.2	1.2	1.2	1.2	1.2
	$T_{G^-G^+t}$	0.0	0.0	0.0	0.0	0.0	0.0
	T_{G^-Tt}	0.6	0.6	0.5	0.6	0.6	0.6
<i>X–H bond scissions</i>							
O–H bond scission	$G^+_{TG^+t}$	104.3	104.9	104.4	105.6	102.1	102.8
	$G^+_{TG^-t}$	104.1	104.8	104.1	105.4	102.2	102.9
C_1 –H bond scission	$G^+_{TG^-t}$	93.6	94.5	93.4	94.9	91.6	92.5
	$T_{G^-G^+t}$	93.6	94.5	93.4	94.9	91.6	92.5
C_2 –H bond scission	$G^+_{TG^-t}$	98.8	99.9	98.8	100.5	97.2	98.3
	$T_{G^-G^+t}$	98.8	99.9	98.8	100.5	96.9	98.1
C_3 –H bond scission	$G^-_{G^+G^-t}$	95.2	96.4	95.5	97.3	93.4	94.6
C_4 –H bond scission	$G^+_{TG^-t}$	100.4	101.5	100.0	101.7	98.2	99.3
C_5 –H bond scission	$G^+_{TG^-t}$	100.5	101.6	100.1	101.8	98.3	99.4
<i>C–X bond scission</i>							
C_1 –O bond scission	G^+_T	93.6	94.7	91.8	93.5	90.6	91.6
C_1 – C_2 bond scission		86.0	86.8	85.1	86.5	83.7	84.5
C_2 – C_3 bond scission	G^+_t	87.6	88.7	87.3	88.9	85.5	86.6
C_3 – C_4 bond scission	TG^+_t	87.1	88.4	85.7	87.8	84.8	86.1

similar hydrocarbons, and are in good agreement with each other having an RMS deviation of $0.5 \text{ kcal mol}^{-1}$ between them. As a result the CBS-QB3 and G3 data sets are used to validate the corresponding input values in the detailed chemical kinetic model.

Consistent with previous studies on *n*-alcohols, the lowest energy configurations, $G^+_{TG^-t}$ and $T_{G^-G^+t}$, have the CCCO in a gauche configuration [12,24,25]. However, iso-pentanol contains a branching methyl group on the γ -carbon that exhibits a strong steric effect on the stability a given configuration. In *n*-pentanol, the difference in relative stability between a gauche and trans CCCO configuration is 0.2 and $0.1 \text{ kcal mol}^{-1}$ for CBS-QB3 and G4, respectively at 0 K [12]. In iso-pentanol, this difference increases to 0.6, 0.5 and $0.5 \text{ kcal mol}^{-1}$ for the CBS-QB3, G3, and G4 methods, respectively, between the lowest energy gauche ($G^+_{TG^-t}$ and $T_{G^-G^+t}$) and trans ($G^-_{G^+Tt}$) configurations. This increase is attributed to long range steric effects between the methyl and hydroxyl substituents that leads to a significant, $1.2 \text{ kcal mol}^{-1}$, destabilizing effect if both are on the same side of the main alkyl chain ($G^+_{TG^+t}$ and $T_{G^-G^-t}$). Similar trends are also seen in the products of the X–H BDE reactions. Table 1 lists the most stable conformers and the X–H and C–X BDE, where X is a C or O (see Table S1 in the Supplementary material for a list of all conformations). In nearly all cases, the trans CCCC conformation, with the branching methyl and OH groups in gauche configurations on opposite sides of the backbone, is the most stable structure. The exception to this is the C_3 –H bond scission (the tertiary hydrogen) product, which takes on a $G^-_{G^+G^-t}$ configuration. Part of the reason for the divergence of this structure from the general trend is that the absence of the hydrogen at this site enables the two methyl groups to occupy positions $\sim 180^\circ$ apart. The resulting gauche CCCC angles are actually $\pm 90^\circ$ not $\pm 60^\circ$.

The α -C–H BDE for iso-pentanol and *n*-pentanol are very similar at 93.6 and $94.0 \text{ kcal mol}^{-1}$, respectively, suggesting that the methyl group has minimal influence on the strength of this bond (Fig. 1) [12]. Similarly, the primary C–H (C_4 –H and C_5 –H) BDE are 100.4 and $100.5 \text{ kcal mol}^{-1}$, respectively, compared to

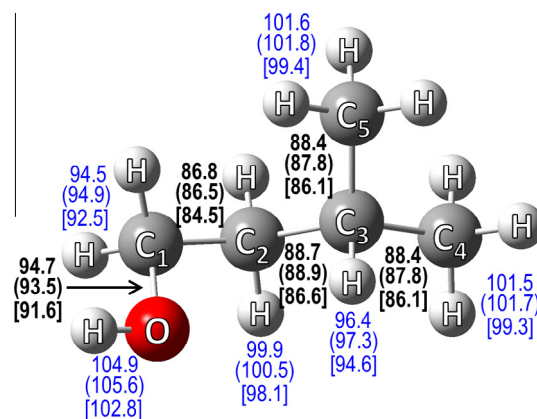


Fig. 1. Schematic diagram of the bond dissociation energies for iso-pentanol. The numbers in the brackets are G4 values, those in the parentheses are G3 values and those without the brackets are the CBS-QB3 values. Blue values are C–H and black are C–C BDEs (kcal mol^{-1} at 298 K).

$100.4 \text{ kcal mol}^{-1}$ in *n*-pentanol. This is consistent with the observation that the effect of the alcohol group on BDE in *n*-pentanol is limited up to the γ -C site. However, unlike *n*-pentanol, the γ -C site in iso-pentanol is a tertiary site with a C–H BDE of $95.2 \text{ kcal mol}^{-1}$, which is only $1.6 \text{ kcal mol}^{-1}$ higher than the α -C–H BDE. This is similar to the analogous site in iso-butanol [11]. In general, the quantum chemistry calculations presented herein are in excellent agreement with BDE values for iso-pentanol estimated by Tsujimura et al. [6] using analogies with butanol isomers. For the C–C BDE, similar to *n*-pentanol, the C_α – C_β in iso-pentanol has the lowest BDE at $86.0 \text{ kcal mol}^{-1}$, which is nearly identical to the corresponding bond in *n*-pentanol, $85.9 \text{ kcal mol}^{-1}$. The remaining C–C BDE values are all higher than the C_α – C_β , though interestingly, the CBS-QB3, G3 and G4 disagree on the extent to which the C_3 – C_4 (and C_3 – C_5), and C_2 – C_3 BDE differ from each other and from C_α – C_β . According to the CBS-QB3 method, the two bond scissions that lead

to the formation of a methyl radical (the C₃–C₄ and C₃–C₅) are 0.5 kcal mol^{−1} lower than the C₂–C₃, 1.1 kcal mol^{−1} above C_α–C_β. According to the G3 data set, the C₃–C₄ and C₃–C₅ BDE are 1.5 kcal mol^{−1} lower than the C₂–C₃ and only 0.6 kcal mol^{−1} above the C_α–C_β. The G4 values place the C₃–C₄ and C₃–C₅ BDE slightly closer to the middle, with values 0.7 kcal mol^{−1} below the C₂–C₃, and 1.1 kcal above the C_α–C_β. In each case the scission of the C–C bond between the tertiary carbon and its adjacent site leads to the formation of a secondary radical site resulting in lower BDE values than those observed in *n*-butanol or *n*-pentanol. As a result, scission of these C–C bonds involving the tertiary site is expected to play a slightly larger role in the overall mechanism of iso-pentanol, and other methylalcohols than it does in *n*-alcohols like *n*-butanol or *n*-pentanol.

2.2. Reaction classes and rate rules

The major classes of elementary reactions considered for the oxidation of iso-pentanol are as follows:

High-temperature reaction classes

1. Unimolecular fuel decomposition.
2. H-atom abstraction from the fuel.
3. Fuel radical decomposition.
4. Fuel radical isomerization.
5. H-atom abstraction reactions from enols (i.e., unsaturated alcohols).
6. Enol-Keto tautomerizations and isomerizations catalyzed by H, HO₂, and formic acid.
7. Addition of H atoms to enols.
8. Enol radical decomposition.
9. Unimolecular decomposition of enols.
10. Reaction of O₂ with α-hydroxypentyl radicals to directly form an aldehyde + HO₂.

Low-temperature reaction classes (R refers to an iso-pentanol radical such as (CH₃)₂C-CH₂CH₂OH and QOOH refers to a hydroxypentylhydroperoxide radical such as (CH₃)₂C(OOH)CH₂CH₂OH)

11. Addition of O₂ to fuel radicals (R + O₂ = ROO).
12. R + ROO = RO + RO.
13. R + HO₂ = RO + OH.
14. R + CH₃O₂ = RO + CH₃O.
15. ROO radical isomerization (ROO = QOOH) including Waddington type reaction mechanism.
16. Concerted eliminations (ROO = enol + HO₂).
17. ROO + HO₂ = ROOH + OH.
18. ROO + H₂O₂ = ROOH + HO₂.
19. ROO + CH₃O₂ = RO + CH₃O + O₂.
20. ROO + ROO = RO + RO + O₂.
21. ROOH = RO + OH.
22. RO decomposition.
23. Formation epoxy alcohols via cyclization.
24. QOOH = enol + HO₂ (radical site beta to OOH group).
25. QOOH = alkene/enol + carbonyl + OH (radical site gamma to OOH group).
26. Addition of O₂ to QOOH (QOOH + O₂ = OOQOOH).
27. Reaction of O₂ with α-hydroxypentylhydroperoxide radicals (e.g., CH₃CH₂CH(OOH)CH₂CH₂OH + O₂).
28. Isomerization of OOQOOH and formation of carbonyl hydroxyalkylhydroperoxide species and OH including Waddington type reactions mechanisms.
29. Decomposition of carbonyl hydroxyalkylhydroperoxide species to form oxygenated radical species and OH.
30. Cyclic oxygenates reactions with OH and HO₂.

As mentioned previously, the CBS-QB3 values for the scission of primary C–H bonds are like values for similar bonds in alkanes, while the values for the OH, α, β, and γ sites are consistent with values for *n*-pentanol, *n*-butanol and iso-butanol. Similar trends exist for C–O and C–C bonds in the iso-pentanol molecule. Therefore, iso-pentanol displays the alkane-like and alcohol-like behavior in specific portions of the molecule, which forms the basis for the allocation of reaction rate constants.

The model developed using standard rate rules for alcohols and alkanes showed discrepancies when predicting low temperature ignition data for iso-pentanol acquired in this study and those available in the literature. To resolve these, the rate constant for reaction class 10 (i.e., O₂ with α-hydroxypentyl radicals to directly form an aldehyde + HO₂) was modified within its uncertainty bounds to better predict ignition delay times at the lowest temperatures. It should be noted that this reaction rate constant draws from previous work on *n*-butanol [11], which has only been tested against a limited set of ignition data. Additionally, the reaction rates for the R + O₂ system for butanol, *n*-pentanol, and iso-pentanol systems have not been studied experimentally or theoretically, so the uncertainties in rate constants and branching ratios of various pathways are rather large. In this study, the high-pressure limit of the rate constant was decreased by a factor of 2.5, while the rate at all other pressures was increased by a factor of 2.5. These adjustments are identical to those previously used in our *n*-pentanol chemical kinetic model [12]. The rate constant for reaction class 27 (O₂ with α-hydroxypentylhydroperoxide radicals) was also modified to keep it in agreement with the rate constant for reaction class 10. Further experimental and modeling studies are needed for these reaction classes to utilize more accurate rate constants in chemical kinetic models. The pressure dependent rate constants for iso-pentanol decomposition from Zhao et al. [8] were utilized in the model, but the rate constant for the reaction leading to the formation of a methyl radical and γ-hydroxybutyl radical was increased by a factor of 10 to improve agreement with the experimental data in shock tubes obtained by Tang et al. [7]. This rate constant was selected for modification based on a sensitivity analysis discussed later.

We also added several reaction pathways based on recent experimental and theoretical studies by Welz et al. [9,10]. The model already includes the Waddington reaction pathway [26,27] involving the β-hydroxypentylperoxy (β-ROO) radical undergoing internal H-atom migration from the OH group via a six-membered transition state ring (i.e., forming an pentylhydroperoxide alkoxy), followed by a rapid decomposition to iso-butanol (3-methylbutanal), formaldehyde, and OH. This reaction is written as a single step irreversible process wherein a beta-ROO radical directly forms the beta scission products (i.e., the reaction forming an pentylhydroperoxidealkoxy intermediate is not explicitly written) with a reaction rate constant of $2.50 \times 10^{10} \exp(-21,886(\text{cal/mol})/RT) \text{ s}^{-1}$. Welz et al. state [9] that similar pathways involving hydrogen transfer from the OH group are important for the γ-ROO and δ-ROO radicals via 7- and 8-membered transition state rings. These reaction pathways were added to the model, and rate constants were applied using logic previously presented by Curran et al. in modeling isomerization reactions in *n*-heptane [16]. We estimate that the ring strain energy decreases by 3 kcal mol^{−1} when going from a 6-membered transition state ring to a 7-membered, and then increases by 3 kcal mol^{−1} when forming an 8-membered transition state ring. The A-factor decreases by a factor of 10 for each incremental increase in ring size because the loss of internal rotors decreases the entropy of the transition state. Welz et al. [10] also presented an unconventional water elimination pathway from hydroxypentylhydroperoxide (QOOH) radicals where the hydroperoxide is on the γ site and the radical is on

the α site. We include this reaction pathway as an irreversible reaction leading to the formation of water, acetone, and vinoxy radical. The rate constant for this reaction is estimated as $1 \times 10^{13} \exp(-20 \text{ (kcal mol}^{-1})/RT) \text{ s}^{-1}$, which is taken from subtracting 10 kcal mol^{-1} from the prescribed activation energy for beta-decomposition of QOOH radicals. This estimate is consistent with the theoretical calculations from Welz et al. [10], which showed an approximately 10 kcal mol^{-1} difference between the barrier height for β -decomposition of QOOH and the water elimination pathway.

2.3. Transport properties

The transport properties for new iso-pentanol related species needed for flame simulations are determined as follows. For stable species, this study uses the correlations developed by Tee, Gotoh, and Stewart [28] as first described by Wang and Frenklach [29] for aromatics, and later by Holley and co-workers for hydrocarbons [30], to calculate the Lennard Jones collision diameter and potential well depth using the critical pressure (p_c), critical temperature (T_c), and acentric factor (ω). The estimation of the acentric factor (ω) is based on Lee-Kesler vapor-pressure relations, which requires the critical pressure (p_c), critical temperature (T_c), and boiling point (T_b) of the species.

The polarizability in cubic Angstroms of stable species is calculated using an empirical correlation [31], which depends on the number of C, H, and O atoms in the molecule. The dipole moment for iso-pentanol is considered identical to that of *n*-butanol, which is a reasonable estimate because polarity is controlled by the hydroxyl functional group and not by the alkyl side chain. The index factor, which describes the geometry of the molecule, is determined from the molecular structure (i.e., 0 for atoms, 1 for linear molecules, and 2 for nonlinear molecules). All C_5 unsaturated oxygenates (e.g. enols, aldehydes, etc.) are assigned transport properties identical to that of *n*-pentanal (i.e., $n\text{-C}_4\text{H}_9\text{CHO}$). For hydroxypentyl radicals and iso-pentenol radical species, the transport properties of their stable counterparts are used.

2.4. Mechanism reduction with DRGX

The method of direct relation graph (DRG) with expert knowledge (DRGX) [32–34] was employed to develop a skeletal mechanism for counterflow diffusion flame simulations. Compared to the original DRG [35] that only allows the specification of a uniform error tolerance for all the species, DRGX eliminates this limitation and allows users to specify species and heat release specific error tolerances (i.e., the expert knowledge, or x -value). Since species and reactions may feature different levels of uncertainties in detailed mechanisms, the species-specific error tolerances render it possible to retain selected important species and associated reaction pathways with higher-than-default accuracy, offering a flexibility to achieve balanced chemical fidelities among different reaction groups. More detailed description on DRGX can be found in [34]. The present reduction with DRGX was performed on reaction states sampled from auto-ignition and perfectly stirred reactor (PSR) in a parameter range that is relevant to the present iso-pentanol counterflow flame experiment. The reduction covers atmospheric pressure, equivalence ratios between 0.5 and 2.0, initial temperature for ignition from 1000–1800 K, and the inlet temperature of 300 K for PSR, which gives reactor temperatures of 1000–2300 K for the pressure and equivalence ratios considered. The codes for DRGX were built on the CHEMKIN-II package [36] and have been revised to account for reactions involving the PLOG keyword in the present detailed model. By specifying the x -values of 0.01 for temperature, 0.1 for H, OH, CH_3 , and iso-pentanol, and

default error tolerance of 0.1, a skeletal mechanism consisting of 187 species and 1237 reactions was obtained.

2.5. Model validation studies

The proposed model for iso-pentanol is validated against a wide range of experimental data covering low-temperature and high-temperature oxidation conditions. The following is a list of specific validation targets presented in this study:

1. Published [7] ignition data from a shock tube at Xi'an Jiaotong University over a temperature range of 1100–1500 K, pressure of 1.0 atm, and equivalence ratios of 0.25, 0.5 and 1.0.
2. Published [6] ignition data from a shock tube (ST) at NUI Galway and RCM at Univ. of Connecticut over a temperature range of 652–1457 K, pressures near 7 atm and 20 atm, and equivalence ratios of 0.5, 1.0 and 2.0 in air.
3. Published [5] concentration profiles from a JSR at CNRS Orleans operating at 10 atm, constant residence time of 0.7 s, equivalence ratio of 1.0, and temperatures from 530 to 1220 K.
4. New ignition data from an RCM at Univ. of Connecticut at pressure near 40 bar, temperature range of 651–776 K, and equivalence ratios of 0.5, 1.0 and 2.0 in air.
5. New ignition data from a shock tube at Rensselaer Polytechnic Institute at equivalence ratios of 0.5, 1.0, and 2.0, temperatures ranging from 819 to 1252 K, and nominal pressures of 40 and 60 bar.
6. New concentration profiles from a JSR at CNRS Orleans operating at 5 atm, constant residence time of 0.35 s, a range of equivalence ratios from 0.35 to 4, and temperatures from 530 to 1220 K.
7. New premixed laminar flame speed and extinction strain rate data from a counterflow flame facility at the Univ. of Southern California.

3. Experimental methodologies

3.1. Shock tube ignition delay

Measurements of ignition delay times were carried out in a heated high-pressure shock tube at Rensselaer Polytechnic Institute using the reflected-shock technique. The shock tube has been previously described by Wang and Oehlschlaeger [37] and references therein, hence, only details relating to the present study are provided here. Ignition delay times were measured for iso-pentanol/air mixtures at equivalence ratios of 0.5, 1.0, and 2.0, at temperatures ranging from 819 to 1252 K, and at nominal pressures of 40 and 60 bar. To maintain sufficient iso-pentanol vapor pressure for the gas-phase experiments the shock tube and reactant mixing vessel were uniformly heated to 130 °C. Reactant mixtures were prepared by direct injection of iso-pentanol into a heated mixing vessel. Following vaporization of the fuel, N_2 and O_2 were added in that order to the mixing vessel from compressed gas cylinders at a molar ratio of 3.76:1; reactant mixture molar fractions were specified via partial pressure. Iso-pentanol was procured from Aldrich at >99% purity and O_2 and N_2 were from Noble Gas Solutions at 99.995% purity. Bursting polycarbonate diaphragms produced shock waves. For experiments with ignition delay times of greater than approximately 1.5 ms, tailored helium/nitrogen driver gases were used for extended reflected-shock test times; for all other experiments pure helium was used as the driver gas.

Ignition delay times were measured behind reflected shocks by monitoring pressure, with a piezoelectric transducer at a side wall

location 2 cm from the end wall, and electronically-excited OH radical chemiluminescence, with a photodetector through the shock tube end wall. Following the passage of the reflected shock, the pressure slowly rises at the test location, due to non-ideal viscous gas dynamics, at a rate of $(dP/dt)(1/P_0) = 1\text{--}3\% \text{ ms}^{-1}$ which is accounted for in simulation of the reflected shock test environment. After an induction period, the reactant mixture ignites and the pressure and OH chemiluminescence signals rise rapidly. The ignition delay time is defined as the time interval between shock reflection at the end wall and the onset of ignition at the end wall, defined by extrapolating the maximum slope in the OH chemiluminescence signal to the baseline. An example experiment is illustrated in Fig. 2 and the measured ignition delay times are appended to the Supplementary material. Reported ignition delay times have estimated uncertainties of $\pm 20\%$ based on the combination of uncertainties in reflected shock temperature and pressure, reactant mixture composition, and determination of the ignition delay from the recorded signals.

3.2. Rapid compression machine ignition delay

The present RCM at the University of Connecticut has been described extensively elsewhere [38]. The configuration is a pneumatically-driven, hydraulically-stopped single piston arrangement. The temperature and pressure at piston top dead center (TDC) are independently adjustable by adjusting the stroke of the piston, TDC clearance, initial pressure and initial temperature. The conditions in the reaction chamber when the piston has reached TDC are referred to as the compressed conditions, represented by the symbols P_c and T_c for pressure and temperature respectively.

Homogeneous fuel and air pre-mixtures are prepared in an approximately 17 L mixing tank. The mixing tank and all tubes and manifolds connecting the tank with the RCM are heated, allowing the study of relatively low vapor pressure fuels. The initial temperature is set above the saturation temperature of iso-pentanol for each mixture studied. The mixing tank is equipped with a magnetic stirrer to ensure complete homogeneity of the mixture.

Prior to mixture preparation, the mixing tank is vacuumed to less than one torr, whereupon liquid fuel (iso-pentanol, Sigma-Aldrich, 99.6% purity) is injected by a syringe through a septum. The syringe is massed before and after the injection, with the difference being the amount of fuel in the mixing tank. Based on this mass,

required proportions of the gaseous oxidizer (O_2 , 99.994% purity, N_2 , 99.999% purity) are calculated. The gases are added to the mixing tank at room temperature and their partial pressures are monitored to ensure the proper mixture concentrations are attained. The partial pressure of the fuel is ensured to be less than the corresponding saturation vapor pressure for all preheat conditions and all equivalence ratios. Finally, the heaters and stirring vane are switched on and the system is allowed approximately 1.5 h to reach steady state.

In this study, the ignition delay in the RCM is defined as the difference in time between the end of compression and the maximum of the time derivative of the pressure. This quantity is indicated graphically in Fig. 3, which shows a representative pressure trace for these experiments. $P(t)$ is the experimental pressure trace and $P'(t)$ is the numerically calculated time derivative of the pressure. The curve labeled “Non-Reactive” is a non-reactive pressure trace whose utility is discussed below. Each condition was repeated at least six times to ensure repeatability. The standard deviation of the ignition delay is less than 10% of the reported value, and the standard deviation therefore represents the current experimental uncertainty. The pressure trace closest to the mean of the scatter is chosen as the indicative pressure trace. Systematic error in the RCM comes primarily from two sources, temperature uncertainty and pressure uncertainty. According to the manufacturers specifications: the uncertainty in the thermocouple measurements of the initial temperature is approximately 2.2 K (between 2% and 5% of the initial temperature); the uncertainty in the initial pressure measurement is approximately 6 torr, which is less than 1% of the initial pressure for all the experiments in this study; the uncertainty of the dynamic pressure measurement during the compression and ignition process is approximately 1%.

The temperature at TDC (T_c) is computed assuming an adiabatic core of gases in the reaction chamber, as detailed by Mittal and Sung [38]. A detailed uncertainty analysis by Weber et al. [39] found that the estimated uncertainty of the temperature calculated by this method is approximately 1%. Further experiments using two-line absorption of water found that the calculated temperatures agreed to within ± 5 K of their measured values [40,41].

To improve the ability of kinetic models to predict the ignition delays measured in the RCM, it is important to account for the heat loss by the reactive gases to the chamber walls. This is done through the application of a time-varying volume profile onto the simulations. The volume profile accounts for the compression

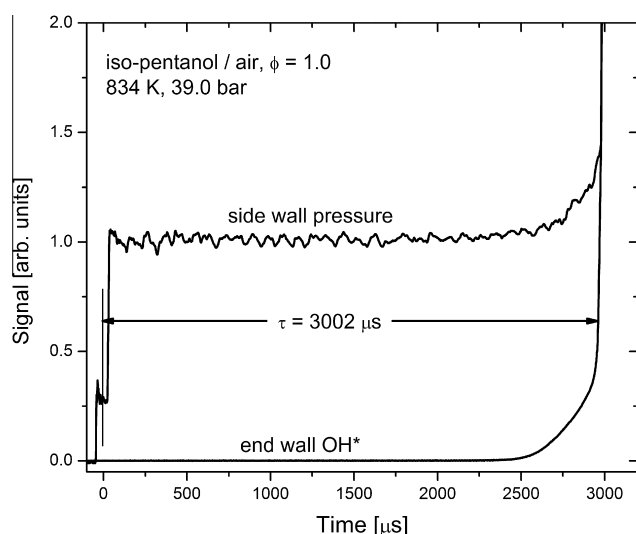


Fig. 2. Example ignition delay time measurement in shock tube for a stoichiometric iso-pentanol/air mixture.

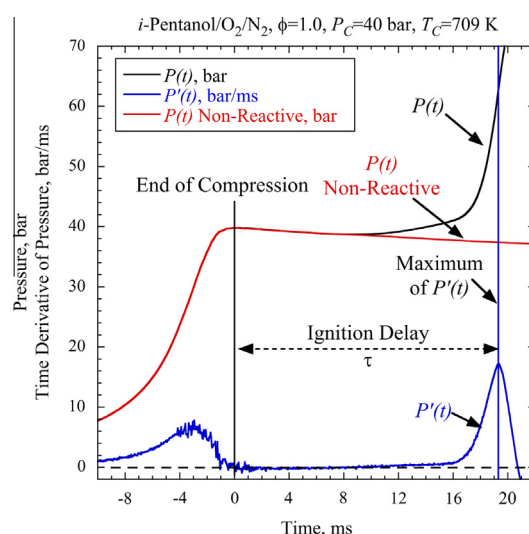


Fig. 3. Example ignition delay time measurement in an RCM for a stoichiometric iso-pentanol/air mixture.

stroke of the piston and models the heat loss after the end of compression as an equivalent volumetric expansion. Such a volume profile is computed from the non-reactive experimental pressure trace by using the specific heat ratio, computed as a function of temperature. A non-reactive experiment is carried out by replacing the oxygen in the oxidizer by nitrogen, to maintain a similar specific heat ratio and eliminate oxidation reactions. The volume profiles are presented in a tabular format in comma-separated value files that can be used directly with CHEMKIN-PRO. The volume profiles and experimental data are available upon request or from the experimentalists' website <http://combdialab.engr.uconn.edu/database/18-rcm-database>.

3.3. Jet stirred reactor experiments

The JSR experimental setup used here has been described earlier [5,42]. The JSR consists of a 4 cm diameter fused silica sphere (42 cm³) equipped with four nozzles of 1 mm i.d.. Before the injectors, the reactants were diluted with nitrogen (<100 ppm H₂O, <50 ppm O₂, <1000 ppm Ar, <5 ppm H₂ from Air Liquide) and mixed. A high degree of dilution (1000–1500 ppm of fuel) was used, reducing temperature gradients and heat release in the JSR. The reactants were high-purity oxygen (99.995% pure from Air Liquide) and high-purity iso-pentanol (>99% pure from Aldrich, which was sonically degassed before use. The reactants were pre-heated before injection to minimize temperature gradients inside the reactor. A HPLC pump (Shimadzu LC10 AD VP) with an on-line degasser (Shimadzu DGU-20) was used to deliver iso-pentanol to an in-house atomizer-vaporizer assembly maintained at 200 °C. Good thermal homogeneity along the vertical axis of the reactor (gradients of <1 K/cm) was measured by thermocouple (0.1 mm Pt–Pt/Rh-10%, located inside a thin-wall fused silica tube). The reacting mixtures were sampled using a movable fused silica low-pressure sonic probe. The samples were directed to analyzers via a Teflon heated line maintained at 200 °C. Analyses were performed online with a FTIR spectrometer (10 m path length, resolution of 0.5 cm^{−1}, 200 mBar in the cell) and off-line after collection and storage in 1 L Pyrex bulbs. Off-line analyses were performed with gas chromatographs (GC) equipped with capillary columns (60 m and 0.32 mm i.d.: DB-624; 50 m and 0.32 mm i.d.: CP-Al₂O₃-KCl; 25 m and 0.53 mm i.d.: Carboplot-P7), a TCD (thermal conductivity detector), and an FID (flame ionization detector). A GC–MS (Varian quadrupole V1200) operating with electron ionization (70 eV) was used for products identification.

Steady state experiments were performed at a constant mean residence time ($\tau = 0.35$ s) and pressure (5 atm), with variable equivalence ratio ($0.35 \leq \phi \leq 4$). The reactants flowed continually in the reactor and the temperature of the gases inside the JSR was increased stepwise. A good repeatability of the measurements and a good carbon balance (typically $100 \pm 10\%$) were observed.

3.4. Counterflow flames

The counterflow flame configuration was used to measure premixed laminar flame speeds and non-premixed extinction strain rates. Experiments were conducted in steady, laminar, and planar counterflow flames at atmospheric pressure and an unburned reactant temperature, $T_u = 353$ K. To measure premixed laminar flame speeds and extinction limits, the counterflow flame was operated in two configurations, Fig. 4a and b, respectively. The diameters of the burners used are 14 mm for flame propagation experiments and 10 mm for extinction limit measurements. The separation distance between the burners was equal to the burner diameter.

The vaporization system consists of a high-precision syringe pump that injects the fuel into a quartz nebulizer. To prevent fuel condensation, the air/fuel mixture delivery line was wrapped with

heating tape and heavily insulated. The partial pressure of the fuel was kept below its vapor (saturation) pressure at all times. The burners were heated using ceramic heating jackets. A K-type thermocouple was used to monitor T_u at the center of the burner exit. It was determined also that the maximum radial variation of T_u was at most 2–3 K. Flow velocities were measured using the digital particle image velocimetry (DPIV) method and the flow was seeded with silicon oil droplets produced by a high efficiency nebulizer.

In order to determine the premixed laminar flame speed, S_u^0 , the twin flame configuration was employed (Fig. 4a). The axial velocity profile along the system centerline just upstream of the flame is first measured. The maximum absolute value of the velocity gradient along the centerline just upstream of the flame is defined as the local strain rate, K , and the minimum point of the axial velocity profile just upstream of the flame is defined as the reference flame speed, $S_{u,ref}$. The laminar flame speed, S_u^0 , is determined by plotting $S_{u,ref}$ against K and non-linearly extrapolating $S_{u,ref}$ to $K = 0$ using a computationally-assisted approach (e.g., [43–45]). Particular care was taken to maintain a plug-flow profile at the nozzle exit due to its relevance to the accuracy of the non-linear extrapolation technique [46].

To determine non-premixed flame extinction strain rates, K_{ext} , a heated fuel/N₂ mixture was counterflowed against an O₂ jet (Fig. 4b). Pure O₂ was chosen as the oxidizer in order to establish flames of small fuel-to-N₂ ratios in the fuel stream. Under such conditions the sensitivity of the flame response to fuel diffusion is augmented, thus allowing for the assessment of both kinetic and molecular transport effects [47]. A flame is established first for a near-extinction condition, and the axial velocity profile along the stagnation stream is measured on the fuel/N₂ stream to determine the prevailing local strain rate, K . Subsequently, the fuel flow rate is slightly modified to achieve extinction. The modification of K due to slight changes in the fuel flow rate has been determined experimentally and numerically to be inconsequential [47–50]. Thus, the value of K measured before the fuel flow rate reduction is defined as the experimentally determined extinction strain rate, K_{ext} . K_{ext} 's were measured on the side of the fuel-containing stream.

Measurements were performed with iso-pentanol (3-methyl-1-butanol) supplied by Alfa Aesar (>99%). Overall, the uncertainties in equivalence ratio and fuel mole fraction, n_{Fuel}/n_{N_2+Fuel} , were estimated to be less than 0.5%. The temperatures of the fuel streams, measured at the center of the nozzle exit, fluctuated within 2–3 °C. The sampling errors in $S_{u,ref}$ and K were determined and their 2σ standard deviations are indicated with uncertainty bars in all figures. More details of the experimental uncertainty analysis can be found in [51]. The axial velocity gradient at the exit of the nozzles, a , was determined experimentally to be between 10 and 50 s^{−1} for all non-premixed extinction measurements. Although the computed K_{ext} exhibits a strong dependence on a [44], the aforementioned measured a are small enough to have a negligible effect on numerical calculations.

4. Results and discussions

4.1. Autoignition delay measurements in RCMs and STs

The model is first validated against high-temperature ST ignition data acquired by Tang et al. [7] over a temperature range of 1100–1500 K, pressure of 1.0 atm, and equivalence ratios of 0.25, 0.5 and 1.0. The experiments were conducted with iso-pentanol/O₂ mixtures highly diluted in Ar. The fuel concentration was maintained at 0.5% and the oxygen concentration was reduced to increase the equivalence ratio from 0.25 to 1.0. The model prediction of ignition delay times for the three equivalence ratios are shown in Fig. 5. Throughout this study, the onset of ignition

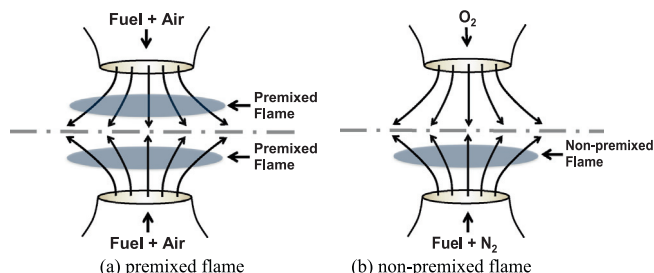


Fig. 4. Counterflow flame configurations used for measuring (a) premixed laminar flame speed and (b) non-premixed flame extinction strain rate.

in constant volume homogeneous batch reactor simulations was determined as the point of maximum temperature rise ($\max dT/dt$), which corresponds closely to the point of maximum pressure rate rise ($\max dP/dt$). Note, that the new pressure dependent rate constants for iso-pentanol decomposition from Zhao et al. [8] were utilized in the model, but the rate constant for the reaction leading to the formation of a methyl radical and γ -hydroxybutyl radical was increased by a factor of 10 to improve agreement with the experimental data. The model well reproduces the experimentally observed trend of increasing ignition delay time with increasing equivalence ratio. This behavior is consistent with that observed in the high-temperature ignition of hydrocarbons at similar conditions (e.g., [52]) where the reactivity is largely driven by the $H + O_2 = OH + O$ chain branching reaction. As the equivalence ratio is increased, the O_2 concentration decreases while the H atom concentration stays relatively constant due to the constant fuel concentration in the reactants. This leads to a decrease of the rate of the chain branching reaction and an increase in ignition delay time. The model also accurately predicts the activation energy of the high-temperature ignition delay curves at all equivalence ratios. Quantitatively, a maximum of 25% difference is observed between simulated and measured ignition delay times, with most points predicted to better than 5% accuracy. Figure 5 also shows that the present model better predicts the experimental activation energy compared to previous models by Dayma et al. [5] and Tsujimura et al. [6].

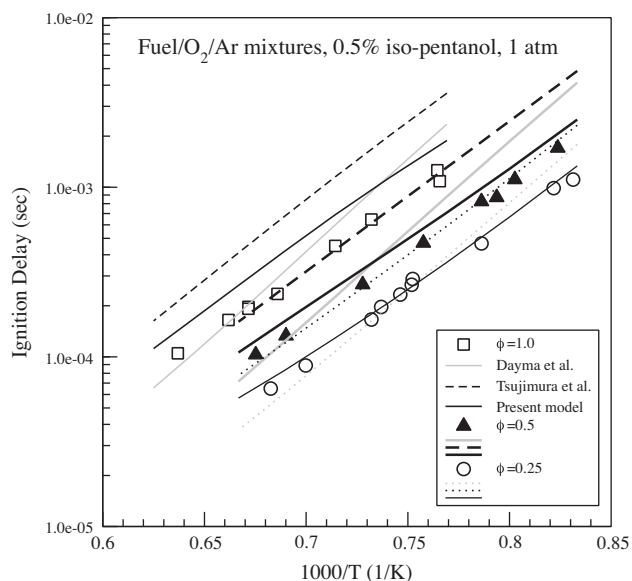


Fig. 5. Ignition delay times for iso-pentanol at 1 atm compared with model prediction.

Figure 6 displays the dominant reaction pathways for the consumption of iso-pentanol at 1300 K, 1 atm, three equivalence ratios, and at 20% fuel consumed. At all equivalence ratios, an important pathway is unimolecular fuel decomposition leading to the formation of methyl radical and the γ -hydroxybutyl radical. This reaction pathway is important for correct prediction of high-temperature ignition delay times because it leads to the formation of methyl radical, and also H radical following eventual β -decomposition of the γ -hydroxybutyl radical via CH_2OH . H-atom abstraction reactions by H radicals are also important in the consumption of iso-pentanol at the given conditions, wherein branching ratios favor the α -site due to its low C–H bond dissociation energy and the primary C–H site due to the high frequency factor (i.e., 6 hydrogen atoms are available for abstraction). Not surprisingly, β -decomposition of the subsequent fuel radicals leads to the formation of smaller radicals and unsaturated species (e.g., alkenes, enols, and/or aldehydes). The importance of the different reaction pathways is relatively insensitive to changes in equivalence ratio. Figure 7 presents a temperature sensitivity analysis at the time of ignition by varying the reaction rate A-factors at 1 atm and 1350 K. This method of sensitivity analysis identifies the same reactions as a brute force sensitivity analysis of reaction rate perturbations on ignition delay time. The sensitivity analysis confirms that the reactivity is largely driven by the $H + O_2 = OH + O$ chain branching reaction. Reactions involving propene, methyl, and propenyl also have a significant effect on iso-pentanol ignition at high temperatures. The most sensitive fuel specific reaction is the decomposition of iso-pentanol leading to the formation of methyl and γ -hydroxybutyl radical. The sensitivity analysis at various equivalence ratios indicates that modifying A-factors will have a greater effect at larger equivalence ratios. For example, the agreement of the model at stoichiometric conditions could be improved without significantly degrading performance at lean conditions by increasing the rate of iso-pentanol decomposition.

Tsujimura et al. [6] measured autoignition data for iso-pentanol/air mixtures using a ST at NUI Galway and an RCM at the University of Connecticut over a temperature range of 652–1457 K, pressure range from 7 to 20 atm and equivalence ratios of 0.5, 1.0 and 2. The present model is validated against the high-temperature ST autoignition experimental data by running homogeneous constant volume simulations, as shown in Figs. 8 and 9 at 7 and 20 atm, respectively, and equivalence ratios of 0.5, 1.0 and 2.0. Overall, the proposed model for iso-pentanol predicts the high-temperature ST ignition data within a factor of 1.5, which is considered an excellent agreement. The experimental data indicates a decrease in ignition delay time as the equivalence changes from lean to rich mixtures, and this trend is well predicted by the model. The trend is different than that observed previously at the higher temperature ST conditions because the reactions driving reactivity are different. Below 1200 K, the combination of $H + O_2 + M = HO_2 + M$ is more dominant than the $H + O_2$ chain branching reaction. In these fuel/air mixtures, the equivalence ratio is increased by increasing the fuel concentration which results in increased radical production, via $\text{fuel} + HO_2 = \text{fuel radical} + H_2O_2$ followed by $H_2O_2 + M = OH + OH + M$, and increased overall reactivity.

The model is also validated against ignition delay times measured in the RCM by running homogeneous constant volume simulations and variable volume history simulations that account for expansion and heat loss in the experiments. For variable volume history simulations, the ignition delay time was taken as the point corresponding to a 400 K increase in post compression temperature. At 7 atm (Fig. 8, gray lines), the model is unable to capture the same activation energy as measured experimentally at low temperatures. The model predictions at the lowest temperatures appear to be in good agreement (i.e., within a factor of 2) with

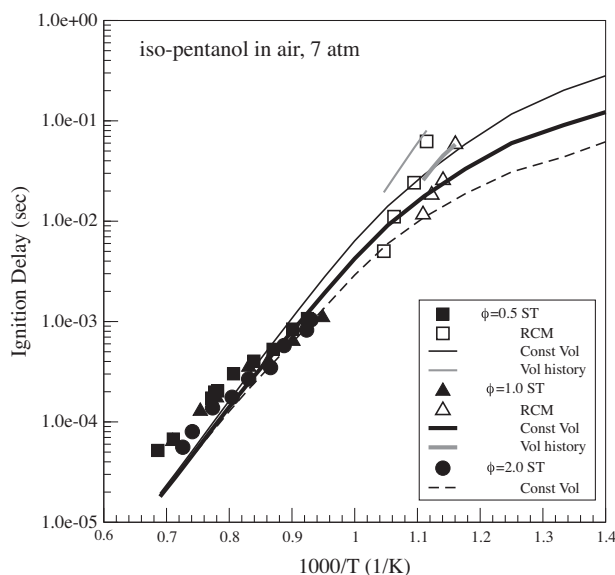


Fig. 8. ST and RCM ignition delay times for iso-pentanol at 7 atm compared with model prediction.

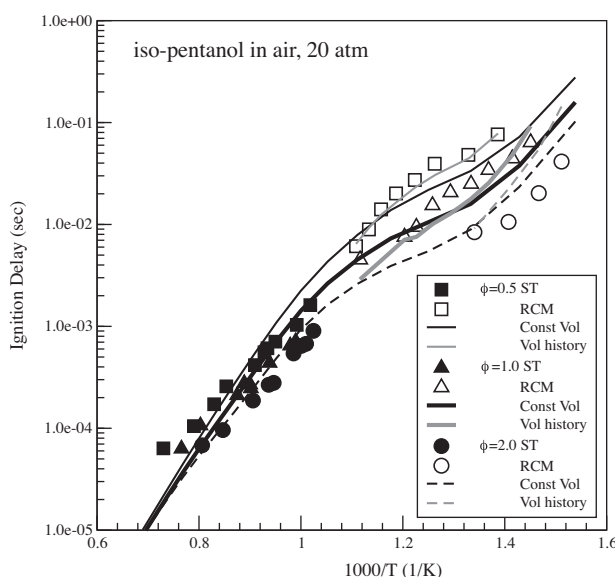


Fig. 9. ST and RCM ignition delay times for iso-pentanol at 20 atm compared with model prediction.

C–H bond at the α site. As discussed previously, subsequent reactions of α -hydroxypentyl with O_2 generate iso-pentanal + HO_2 , an OH terminating pathway. The other hydroxypentyl radicals tend to add to molecular oxygen and form hydroxylalkylperoxy (ROO) radicals. These radicals are mainly isomerized to hydroxylalkylhydroperoxide (QOOH) or undergo a 5-centered concerted reaction to form HO_2 and an enol species [53]. Around 18% of β -hydroxylalkylperoxy radicals are decomposed to produce iso-butanal (2-methylpropanal), formaldehyde, and OH radical via ROO isomerization and β -scission reactions by the Waddington mechanism [26,27]. It is also interesting to note that a similar pathway involving hydrogen transfer from the OH group is shown to be important for the γ -ROO (~6%) via a 7-membered transition state ring, which is a reaction sequence we included based on the work of Welz et al. [9]. More discussions of low temperature reaction pathways are presented in the JSR section below.

The model was further validated against new ignition delay data in RCMs and STs near 40 bar and 60 bar, as shown in Figs. 12 and 13. As previously observed in STs at higher temperatures and lower pressures, the model accurately predicts the ST ignition delay data above 900 K at both 40 bar and 60 bar. Several data points were measured in the shock tube from 819 to 900 K. The model is slightly less predictive in this region, with differences of up to a factor of two. Ignition delay data near 40 bar and temperatures ranging from 651 to 776 K were acquired using an RCM and are plotted together with the ST data. The ignition data in the RCM and ST are in good qualitative agreement, displaying the expected decrease in ignition delay with increasing temperature.

The model well predicts the observed trend of decreasing ignition delay time with increasing equivalence ratio, which occurs because a higher fuel concentration results in greater radical production at these conditions. Constant volume and volume history simulations at the 40 bar RCM conditions (Fig. 12) indicate the model can well predict ignition delay times at stoichiometric conditions, but cannot well predict RCM ignition delay data at lean and rich conditions. The difference (i.e., spread) in ignition delay times across various equivalence ratios is similar to that observed for other alcohols in the same facilities (e.g., *n*-butanol at 15 bar [54] and tert-butanol at 30 bar [39]). The primary issue with the model is its phi-sensitivity; predicted ignition delay times need to be increased at lean conditions yet decreased at rich conditions, implying that the system's reactivity is controlled by different phenomenon at different equivalence ratios. At these low temperatures the model's reactivity is driven by the overall peroxy reaction sequence, $R + O_2 = ROO = QOOH(+O_2) = OOQOOH = 2OH + \text{products}$, including the inhibitive direct (i.e., concerted) HO_2 elimination and QOOH decomposition routes. An increase (or decrease) in any reaction rate constant along this reaction sequence will move the reactivity of the system in the same direction at all equivalence ratios. Therefore, we were unable to identify a single reaction rate constant modification that would decrease overall reactivity at lean conditions while increase it at rich conditions.

Representative experimental and simulated pressure profiles for the lean, stoichiometric, and rich conditions are shown in

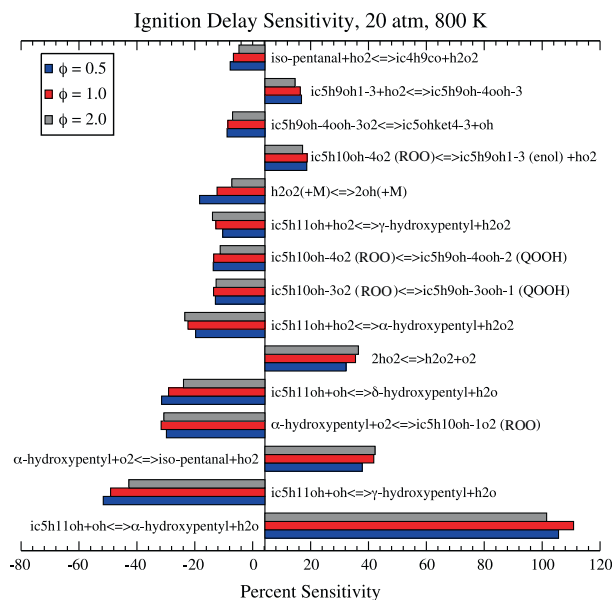


Fig. 10. Sensitivity of the ignition delay to changes in the reaction rate coefficients for three equivalence ratios. Initial conditions for constant-volume adiabatic simulations are 800 K and 20 atm.

Fig. 14a–c respectively, with the profiles shifted so TDC occurs at 0 s. Interestingly, the experimental pressure traces after the induction period do not show a sharp increase in pressure (i.e., heat release is more gradual, similar to two-stage ignition). For the lean case, there is a moderate heat release 25 ms after TDC followed by a larger heat release event, and similar behavior is observed at other equivalence ratios. It is noted that similar heat release prior to the main ignition event was found in an HCCI engine experiment using iso-pentanol by Yang et al. [3] and was termed Intermediate Temperature Heat Release (ITHR) in their work.

The pressure profile of the present simulations qualitatively agrees with the experimental data, in that the simulated pressure traces deviate from the non-reactive trace prior to the main ignition event, although the ignition delay itself does not necessarily agree very well. For lean and stoichiometric cases the simulated ignition delay times are fast compared to the data, whereas at rich conditions they are too slow.

A sensitivity analysis of the ignition delay to changes in the reaction rate coefficients for initial conditions of 40 atm and 689 K, for three equivalence ratios, is shown in Fig. 15. As before, positive sensitivity indicates that increasing the rate coefficient of that reaction increases the ignition delay. Similar to the 20 atm sensitivity analysis, the most sensitive reactions are the H-abstractions from the fuel and the subsequent reactions of these initial fuel radicals. However, stronger equivalence ratio dependence of the sensitivity results is seen in Fig. 15 as compared to Fig. 10. It is interesting to note that the most sensitive reaction, formation of the α -hydroxypentyl radical through H-abstraction by OH, is nearly twice as sensitive at $\phi = 0.5$ than at $\phi = 2.0$, while most of the other reactions have nearly the same sensitivity for the three equivalence ratios. In addition, the reaction of formaldehyde and hydroxyl radical to form formyl radical and water shows

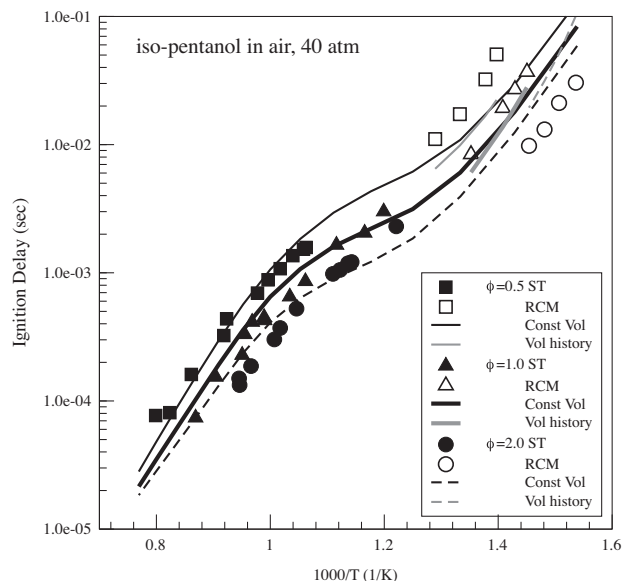


Fig. 12. ST and RCM ignition delay times for iso-pentanol near 40 bar compared with model prediction.

up as somewhat sensitive, especially for the lean case. The importance of this reaction is demonstrated by the path analysis shown in Fig. 11 (similar results are obtained for a path analysis at 40 atm as at 20 atm). Formaldehyde is a significant product in the decomposition of the β - and γ -hydroxypentyl radicals, as well as the pentoxy radical. Furthermore, the reaction of two hydroperoxyl molecules to form hydrogen peroxide and oxygen molecule is the

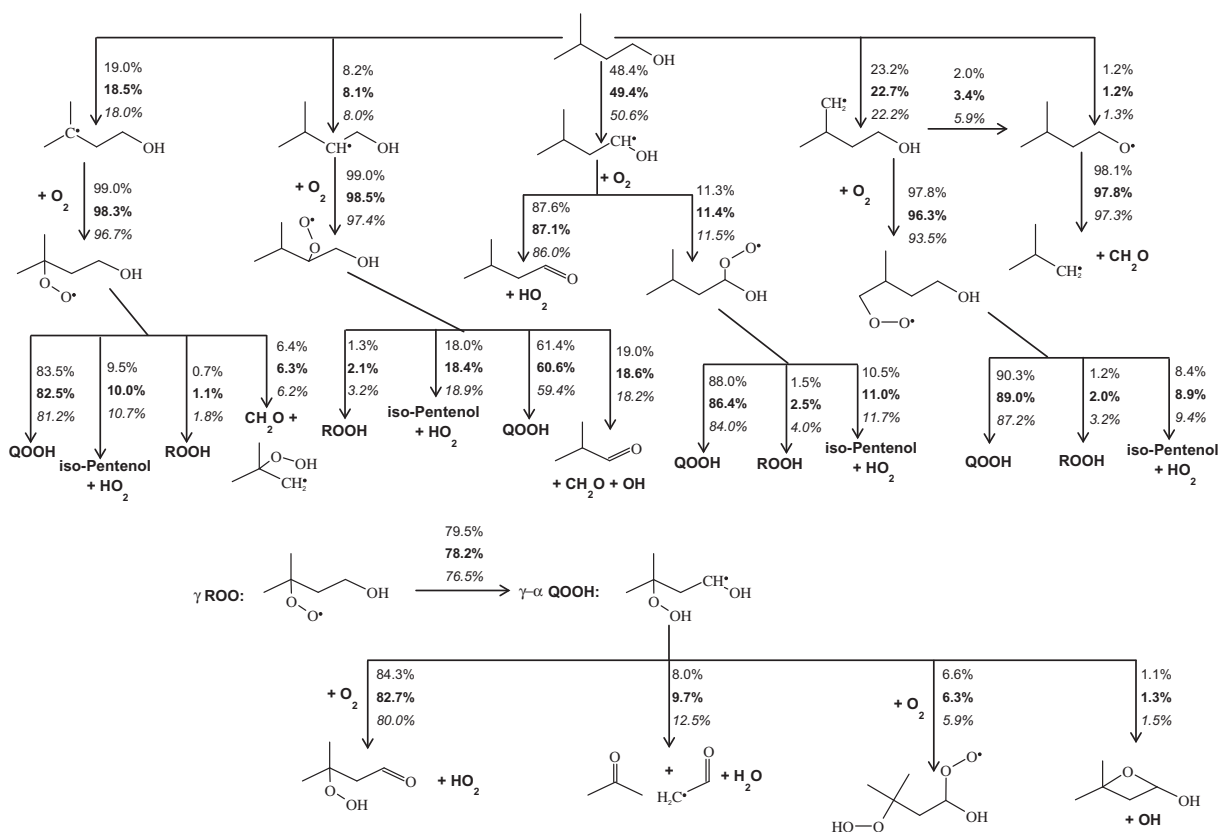


Fig. 11. Reaction path analysis for iso-pentanol at 800 K, 20 atm, and three equivalence ratios, based on constant-volume adiabatic simulations. Plain text: $\phi = 0.5$. Bold text: $\phi = 1.0$. Italic text: $\phi = 2.0$. The flux is integrated up to 20% fuel consumption.

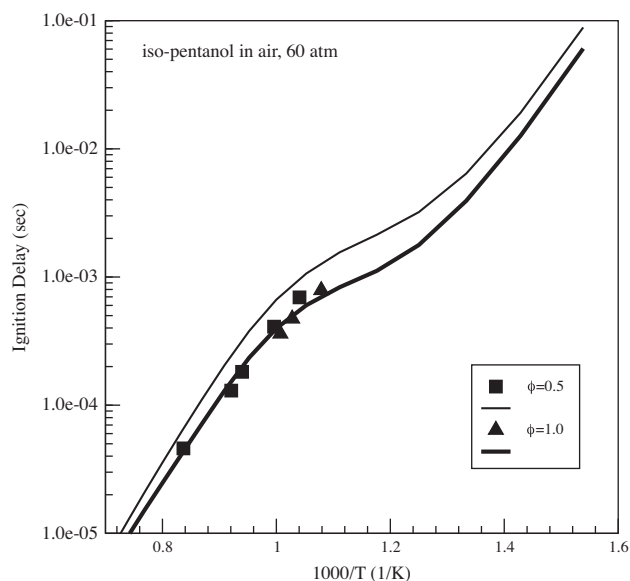


Fig. 13. ST ignition delay times for iso-pentanol near 60 bar compared with model prediction.

seventh most sensitive reaction. This reaction is important as it releases the most heat during the ITHR period prior to the main ignition for all three equivalence ratios and because the rapid reaction of the α -hydroxypentyl radical to form iso-pentanal and hydroperoxyl is important in alcohol combustion. In view of the equivalence ratio dependence shown in Fig. 15, these sensitivity analysis results suggest that it may be possible to adjust multiple reaction rates in the low temperature chain branching pathways to decrease reactivity at lean conditions but increase reactivity at rich conditions, which warrants further investigation.

4.2. Jet-stirred reactor measurements

Dayma et al. [5] measured concentration profiles of iso-pentanol oxidation in a jet-stirred reactor (JSR) at 10 atm, constant residence time, τ , of 0.7 s over a range of equivalence ratios (0.35–4) and temperatures from 530 to 1220 K. The present model is validated against the JSR data for an equivalence ratio of 1.0, as previous studies [55,56] have shown that validation at these conditions is representative of model predictability at other equivalence ratios. Figure 16 compares the mole fractions of reactants (iso-pent-

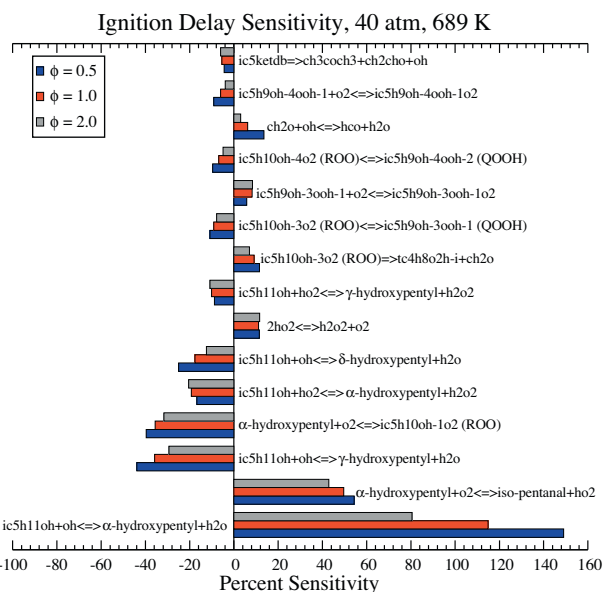


Fig. 15. Sensitivity of the ignition delay to changes in the reaction rate coefficients of the given reactions, for conditions of 40 atm, 689 K, and three equivalence ratios.

anol and O_2), major products (CO , CO_2 , H_2 and H_2O) and intermediate species (methane, ethylene, propene, iso-butene, formaldehyde, etc.). The predictions show good agreement with experimental data; however, the simulation results reveal that the fuel consumption occurs at slightly lower temperature (~ 50 K) than the measurement.

Figures 17–21 present new experimental data for iso-pentanol in the JSR at 5 atm and equivalence ratios of 0.35, 0.5, 1.0, 2.0, and 4.0, respectively. Experimental profiles are presented for the reactants (iso-pentanol and O_2) and the important products. As seen in the figures, the consumption of the fuel occurs above 800 K. Experiments carried out in the temperature range of 550–770 K showed no cool-flame oxidation for the most reactive mixture ($\phi = 0.35$) under the present experimental conditions. The conversion of the fuel is accompanied with a rapid formation of iso-pentanal/3-methyl-1-butenol, formaldehyde, iso-butene, and acetaldehyde/ethanol. Methane, ethylene, and propene are the main intermediate hydrocarbons formed. Formaldehyde, iso-pentanal/3-methyl-1-butenol, acetaldehyde/ethanol, iso-butanal/2-methyl-1-propenol, and 2-propenal are the main oxygenated products formed during iso-pentanol oxidation. The experimental

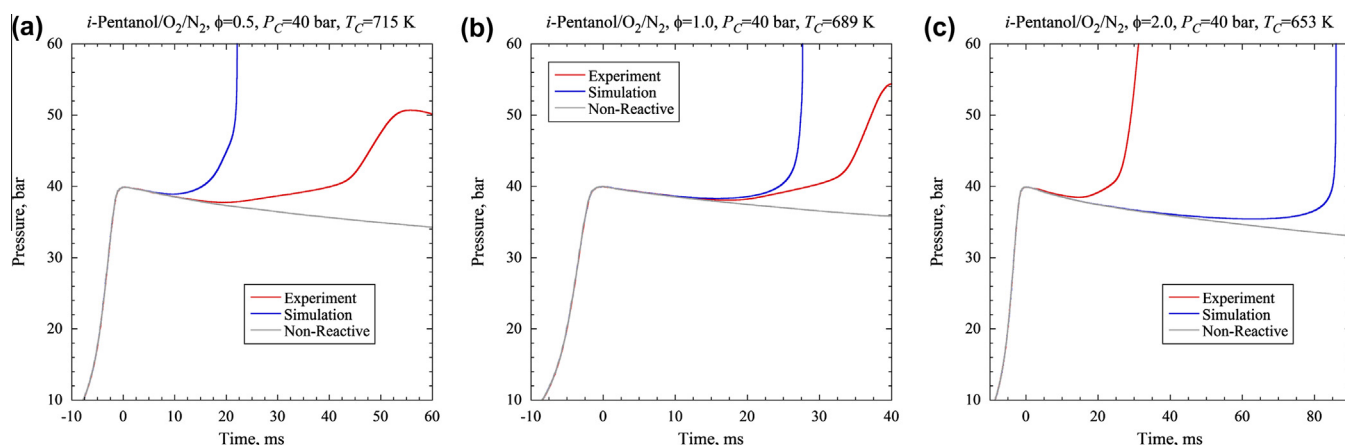


Fig. 14. Experimental reactive (red), simulated reactive (blue), and experimental non-reactive (gray) pressure profiles at 40 bar for (a) $\phi = 0.5$, (b) $\phi = 1.0$, (c) $\phi = 2.0$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

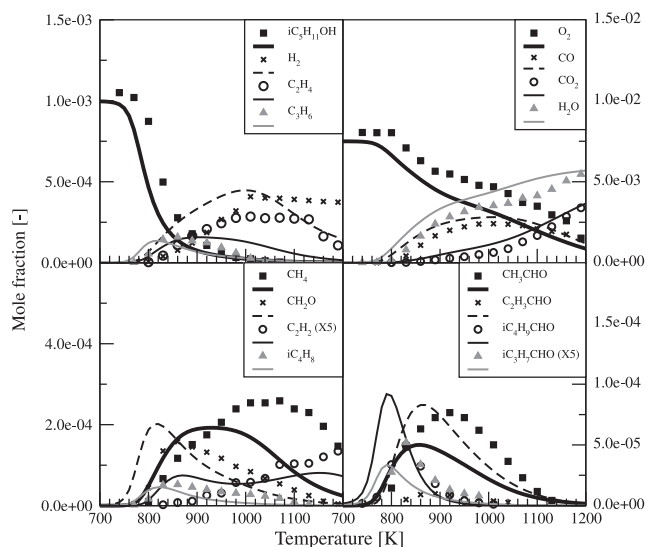


Fig. 16. iso-Pentanol oxidation in a JSR at 10 atm, $t = 0.7$ s and $\phi = 1.0$. The initial fuel mole fraction was 0.1%. Experimental data [5] (symbols) are compared to calculations (lines).

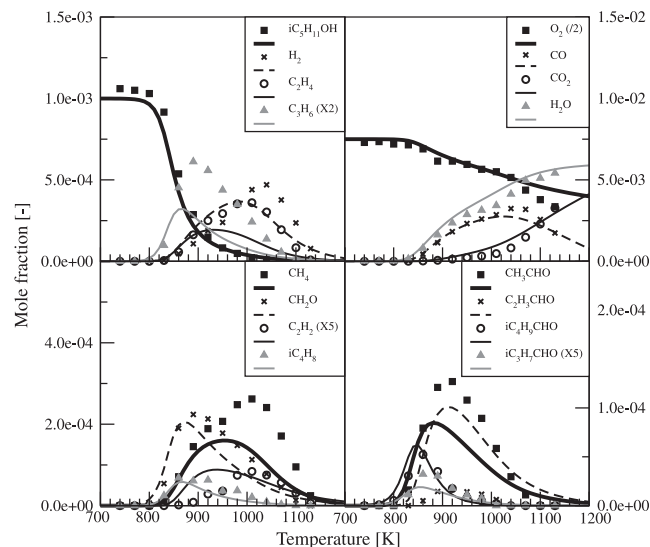


Fig. 18. iso-Pentanol oxidation in a JSR at 5 atm, $t = 0.35$ s and $\phi = 0.50$. The initial fuel mole fraction was 0.1%. Experimental data (symbols) are compared to calculations (lines).

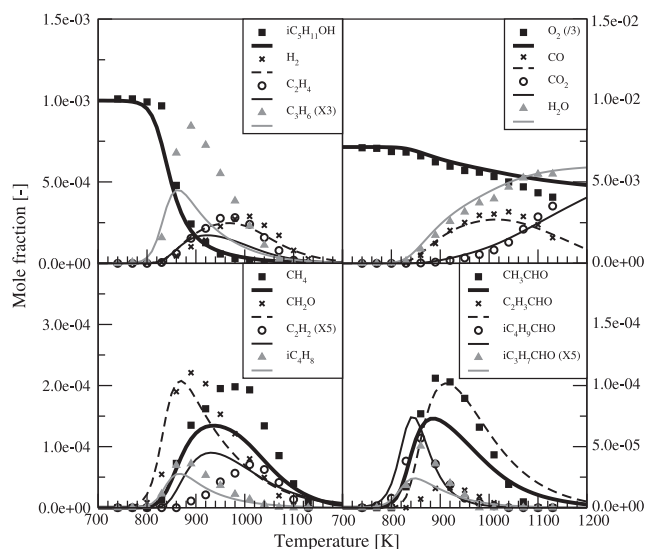


Fig. 17. iso-Pentanol oxidation in a JSR at 5 atm, $t = 0.35$ s and $\phi = 0.35$. The initial fuel mole fraction was 0.1%. Experimental data (symbols) are compared to calculations (lines).

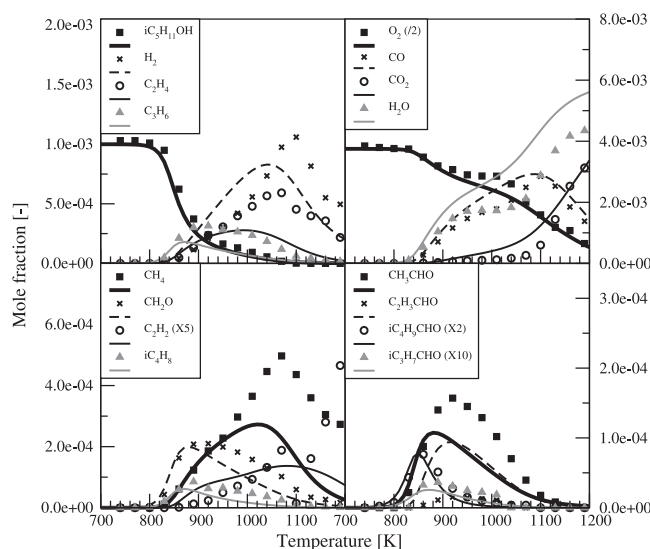


Fig. 19. iso-Pentanol oxidation in a JSR at 5 atm, $t = 0.35$ s and $\phi = 1.0$. The initial fuel mole fraction was 0.1%. Experimental data (symbols) are compared to calculations (lines).

results presented herein are consistent with those obtained at higher pressures [5], so readers are referred to [5] for a more comprehensive discussion of iso-pentanol oxidation in the JSR.

Model predictions at 5 atm JSR conditions are also presented for various equivalence ratios. The model's ability to reproduce the experimental data is discussed qualitatively and quantitatively. The model's prediction is considered good if the shape of the model profile closely matches the experimental profile, and if the predicted maximum mole fraction is within a factor 2 of the measured maximum mole fraction. The concentration of the reactants of iso-pentanol and O_2 is well predicted at all temperatures, and so is the concentration of the major product species CO, CO_2 , and H_2O . The model well predicts the minor species CH_4 , H_2 , C_2H_2 , CH_2O , and iso-butene. The shape of C_2H_4 and C_3H_6 profiles are well reproduced, but their maximum concentrations are under predicted

by more than a factor of 2. We present the model's summed prediction of ethanal + ethenol, 2-methylpropanal + 2-methyl-1-propenol, and iso-pentanal + 3-methyl-1-butenol because surface tautomerization reactions render these species indistinguishable. The concentration of iso-pentanal + 3-methyl-1-butenol and ethanal + ethenol are well predicted; however, 2-propenal is over predicted and 2-methylpropanal + 2-methyl-1-propenol is under predicted within a reasonable range. The formation of the $C_5H_{10}O$ isomers (aldehydes/enols) in the JSR is attributed primarily to reactions of α -hydroxypentyl reactions with O_2 to form an aldehyde.

Figure 22 presents a reaction path analysis for iso-pentanol oxidation in the JSR at 850 K, 5 atm, and three equivalence ratios. Hydrogen abstraction reactions by OH radicals are the most important, with abstraction from the α , γ , and primary sites being the most important. The fuel radicals are primarily consumed via

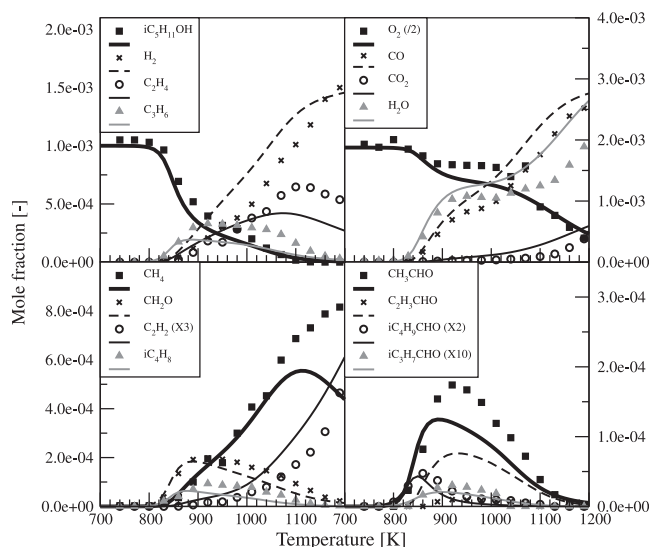


Fig. 20. iso-Pentanol oxidation in a JSR at 5 atm, $t = 0.35$ s and $\phi = 2.0$. The initial fuel mole fraction was 0.1%. Experimental data (symbols) are compared to calculations (lines).

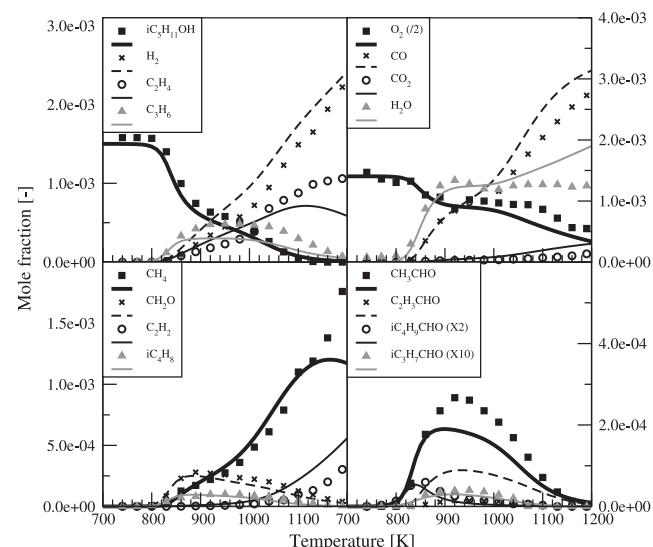


Fig. 21. iso-Pentanol oxidation in a JSR at 5 atm, $t = 0.35$ s and $\phi = 4.0$. The initial fuel mole fractions was 0.1%. Experimental data (symbols) are compared to calculations (lines).

reactions with O_2 at lean and stoichiometric conditions, but at rich conditions ($\phi = 4.0$) β -decomposition reactions become important due to the lack of O_2 . In general, the low temperature reaction pathways are similar to those presented and discussed previously in Fig. 11. As shown earlier, hydrogen transfer from the OH group is shown to be non-negligible for the β -ROO [26,27] and γ -ROO [9]. The fate of α -hydroxypentylhydroperoxide radical is also presented in Fig. 11 and Fig. 22, with reactions leading to the formation of a cyclic ether plus OH and a concerted elimination to carbonylhydroperoxide plus HO_2 depicted. The hydroperoxide

pentaldehyde formation is dominant under ignition conditions (Fig. 11) and decomposes to give back OH. Therefore, this reaction path is a form of low temperature chain branching which produces one OH and one HO_2 radical. It is interesting to note that the α -hydroxypentylhydroperoxide radical with a hydroperoxy group on the tertiary site could add another O_2 and isomerize to form a carbonyl hydroxyalkylhydroperoxide plus OH, but this reaction sequence is not favored because the 6-member ring internal H-atom migration reactions are not possible due to the presence of a methyl branch. The new path proposed by Welz et al. [10]

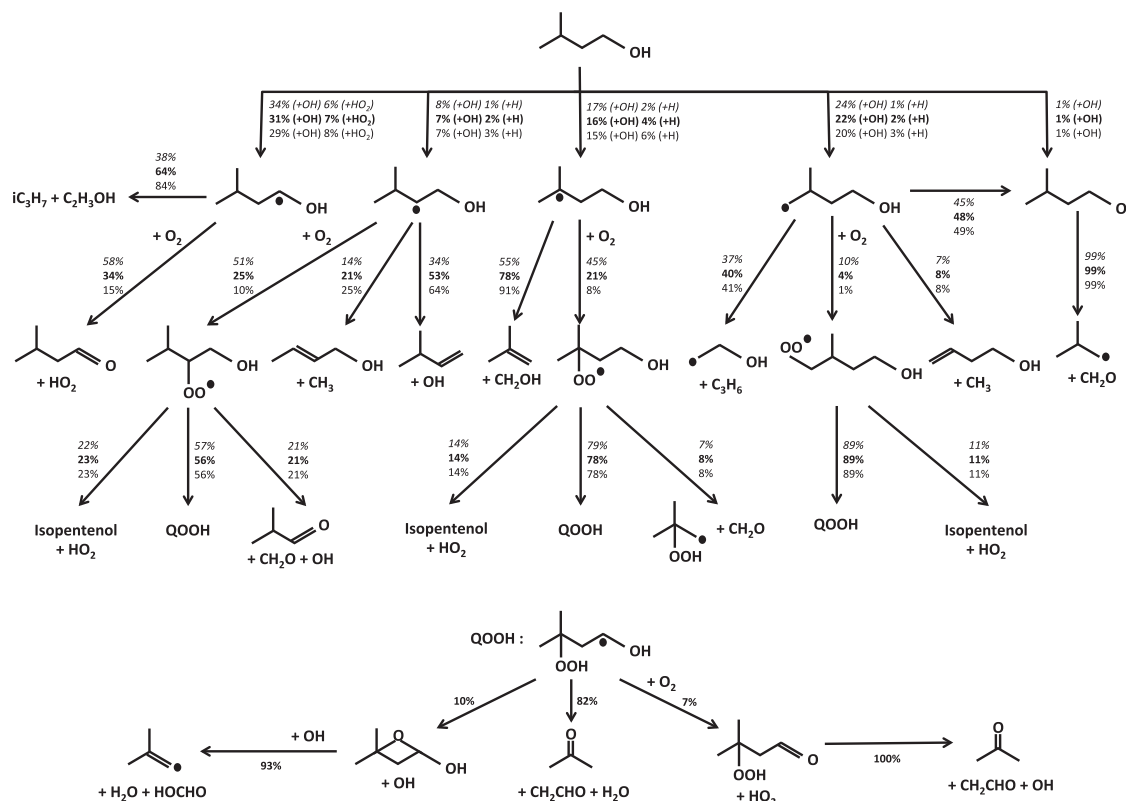


Fig. 22. Reaction path analysis for iso-pentanol oxidation in a JSR at 850 K, 5 atm, $\phi = 0.35$ (italic), $\phi = 1.0$ (bold) and $\phi = 4.0$ (plain).

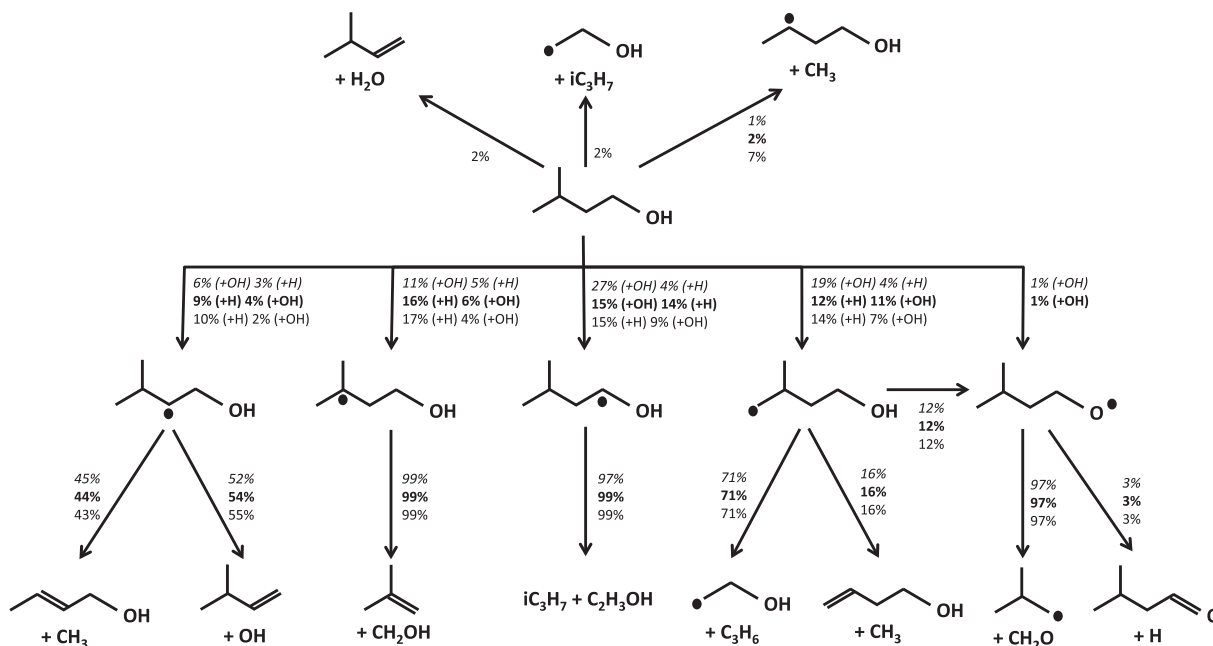


Fig. 23. Reaction path analysis for iso-pentanol oxidation in a JSR at 1050 K, 5 atm, $\phi = 0.35$ (italic), $\phi = 1.0$ (bold) and $\phi = 4.0$ (plain).

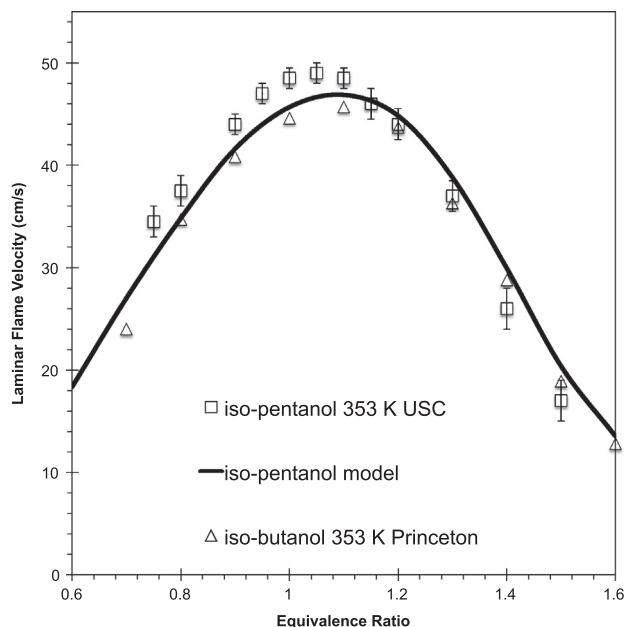


Fig. 24. Laminar flame speeds at unburnt gas temperature of 353 K. Lines show model predictions for iso-pentanol. Experimental data for iso-butanol data [57] is also presented.

involving water elimination from QOOH produced from the γ -peroxy radical was included, and it is shown to be the dominant pathway for the consumption of this QOOH radical under JSR conditions (Fig. 22).

Figure 23 presents a reaction path analysis for iso-pentanol oxidation in the JSR at 1050 K, 5 atm, and three equivalence ratios. In general, the important reactions are similar to those observed previously in Fig. 6 for the high temperature (1300 K) ST simulations, except the role of hydrogen abstraction and β -decomposition are more important in the JSR due to the lower temperatures and

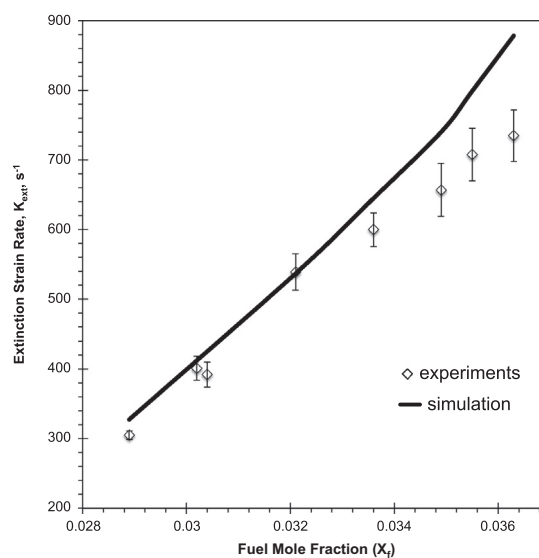


Fig. 25. Extinction strain rates measured for iso-pentanol counterflow flames.

perfect mixing during the entire residence time. OH and H radicals are the primary abstractors with OH radical being dominant at lean conditions and H radical being more important at stoichiometric and rich conditions.

4.3. Counterflow flame measurements and predictions

New experimental data in counterflow flames was acquired for premixed laminar flame speeds and extinction strain rates of non-premixed flames, as shown in Figs. 24 and 25, respectively. The skeletal model generated using DRGX was used to simulate extinction of these one-dimensional flames. Premixed flame simulations utilized the detailed high temperature mechanism without reduction. The model under predicts laminar flame speeds of

iso-pentanol at equivalence ratios below 1.2, while well predicting them at richer conditions. It is interesting to note that the iso-pentanol model predicts flame speeds similar to the measured data for iso-butanol at the same unburnt gas temperature of 353 K [57]. Previous model predictions of iso-butanol [11] show similar agreement with this experimental data. Therefore, the models for iso-pentanol and iso-butanol suggest that the two fuels have similar flame speeds, which is due to the fact that both models are developed using similar reaction classes and rate rules. This consistency of flame speeds for fuels of similar structure but different chain length has been shown experimentally and computationally for *n*-alkanes [44]. The model also well predicts extinction strain rates at lower fuel mole fractions, but over predicts them at higher fuel mole fractions, as shown in Fig. 25.

5. Conclusions

In summary, a new model for iso-pentanol was validated against new and existing experimental data in shock tubes, rapid compression machines, jet stirred reactors, premixed flames, and non-premixed flames. The model was developed using methods established previously for the butanol isomers and *n*-pentanol, as well as recently published fundamental studies on iso-pentanol oxidation. Bond dissociation energies in iso-pentanol were calculated using *ab initio* methods to give insights into the fuel's initial decomposition and aid in the estimation of rate constants. The calculations indicate that iso-pentanol displays alkane-like and alcohol-like moieties, which is useful information for initial allocation of rate constants. Several modifications to important reaction rate constants were made to improve agreement against a broad set of experimental data.

Shock tube ignition delay times were measured for iso-pentanol/air mixtures at equivalence ratios of 0.5, 1.0, and 2.0, at temperatures ranging from 810 to 1252 K, and at nominal pressures of 40 and 60 bar. New jet stirred reactor experiments are reported at 5 atm and four different equivalence ratios. Rapid compression machine ignition delay data was obtained at 40 bar, three equivalence ratios, and temperatures below 800 K. Premixed laminar flame speeds and non-premixed extinction strain rates were obtained using the counterflow configuration, and the method of DRGX was employed to eliminate unimportant species and reactions in the detailed iso-pentanol mechanism for prediction of non-premixed flame behaviors.

The present model shows reasonable overall agreement with the data obtained from a wide variety of experimental conditions. The most important path for consumption of fuel radicals at low and intermediate temperatures was the reaction of the α -hydroxypentyl radical with O_2 to form iso-pentanal and HO_2 , a path that does not contribute to the low temperature branching. However, sufficient low temperature chain branching involving the γ and δ fuel radicals occurred in the model that it was able to reasonably reproduce low temperature ignition and reactivity observed in the experiments. The model included reaction pathways from recent low temperature oxidation studies of iso-pentanol [9,10]. Reaction path analysis showed that hydrogen transfer from the OH group by the γ -ROO radical was small but non-negligible, while the path involving water elimination from QOOH produced by the γ peroxy radical considerably altered the understanding of how this radical is consumed. Sensitivity analysis showed that neither of these reaction pathways is expected to significantly alter iso-pentanol low temperature ignition characteristics under the conditions studied here.

In the RCM, measured pressure histories showed an interesting behavior of a slow initial pressure rise prior to an increased pressure rise during ignition. The slow initial pressure rise may be

attributed to the role of the Waddington mechanism [26,27] in consuming the fuel via the production and re-cycling of OH radicals during the pre-ignition phase, but the model was not able to reproduce this behavior. The effects of equivalence ratio on RCM ignition delay were not well captured by the model, and further investigation into reaction rate constants for various low temperature oxidation reactions need to be studied experimentally and/or theoretically.

The present study demonstrates that chemical kinetic reaction models can be developed using a consistent set of reaction classes and rate rules. This approach generates models that qualitatively well predict a wide range of combustion data. Quantitative predictions are also accurate at the majority of conditions studied, but benefits can be gained from further research on alcohol specific reaction mechanisms and rate constant estimates, primarily at pressures greater than 10 atm and temperatures in the range 550–950 K. These are also the environmental conditions of relevance to the low temperature combustion (LTC) engines currently being developed to utilize novel fuel technologies. Therefore, comprehensive experimental and modeling studies on candidate next-generation biofuels, such as presented herein for iso-pentanol, can aid in the development of future fuels for future engines.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.06.022>.

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